

Aflevering 5

Wednesday, 13 November 2024 07:56

Exercise 7: (Exercise 3 from Exam of May 2023) For a function $f: \mathbb{R} \rightarrow \mathbb{C}$ we consider the Fourier Transform

$$\hat{f}(\lambda) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\lambda x} dx$$

that is defined for any $\lambda \in \mathbb{R}$ for which the integral above converges.

Take $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) := e^{-|x|} \sin(x), \quad \text{for } x \in \mathbb{R}.$$

1. With the help of Euler's formulas show that the Fourier Transform $\hat{f}(\lambda)$ of the above considered function f is given by

$$\hat{f}(\lambda) = \frac{i}{\sqrt{2\pi}} \left(\frac{1}{1 + (\lambda + 1)^2} - \frac{1}{1 + (\lambda - 1)^2} \right).$$

Euler's formula shows that

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

Which gives us $f(x) = e^{-|x|} \frac{e^{ix} - e^{-ix}}{2i}$

$$\hat{f}(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\lambda x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{2i} \int_{-\infty}^{\infty} \left(e^{-|x|+ix} - e^{-|x|-ix} \right) e^{-i\lambda x} dx$$

We split it up in $x < 0$ where $e^{-|x|} = e^x$
and $x \geq 0$ where $e^{-|x|} = e^{-x}$

$$= \frac{-i}{2\sqrt{2\pi}} \left(\int_{-\infty}^0 e^{(1-i(\lambda-1))x} dx - \int_{-\infty}^0 e^{(1-i(\lambda+1))x} dx + \int_0^{\infty} e^{(-1-i(\lambda-1))x} dx - \int_0^{\infty} e^{(-1-i(\lambda+1))x} dx \right)$$

$$= \frac{-i}{2\sqrt{2\pi}} \left(\frac{1}{1-i(\lambda-1)} - \frac{1}{1-i(\lambda+1)} + \frac{-1}{-1-i(\lambda-1)} - \frac{-1}{-1-i(\lambda+1)} \right)$$

$$= \frac{i}{\sqrt{2\pi}} \left(\frac{1}{1+(\lambda+1)^2} - \frac{1}{1+(\lambda-1)^2} \right)$$

1/3

2. Show that $\hat{f}(\lambda)$ is an odd function, and from that derive the value of the following integral, for every $x \in \mathbb{R}$,

$$I(x) = \int_{-\infty}^{\infty} \left(\frac{1}{1+(\lambda+1)^2} - \frac{1}{1+(\lambda-1)^2} \right) \sin(\lambda x) d\lambda.$$

For $\hat{f}(\lambda)$ to be odd we need $\hat{f}(-\lambda) = -\hat{f}(\lambda)$

$$\begin{aligned} \hat{f}(-\lambda) &= \frac{i}{\sqrt{2\pi}} \left(\frac{1}{1+(-\lambda+1)^2} - \frac{1}{1+(-\lambda-1)^2} \right) \\ &= \frac{i}{\sqrt{2\pi}} \left(\frac{1}{1+(\lambda-1)^2} - \frac{1}{1+(\lambda+1)^2} \right) \\ &= -\hat{f}(\lambda) \end{aligned}$$

Since $\hat{f}(\lambda)$ is odd we from Theorem 2.1

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\lambda) e^{i\lambda x} d\lambda = \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} \hat{f}(\lambda) \cos(\lambda x) d\lambda + \int_{-\infty}^{\infty} i \hat{f}(\lambda) \sin(\lambda x) d\lambda \right) \\ &= \frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\lambda) \sin(\lambda x) d\lambda \\ &= \frac{i}{\sqrt{2\pi}} \frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{1+(\lambda+1)^2} - \frac{1}{1+(\lambda-1)^2} \right) \sin(\lambda x) d\lambda \end{aligned}$$

$$\Rightarrow I(x) = -2\pi f(x)$$

3. Calculate the value of the integral

$$J = \int_{-\infty}^{\infty} \left(\frac{1}{1+(\lambda+1)^2} - \frac{1}{1+(\lambda-1)^2} \right)^2 d\lambda.$$

(Hint: Use $(\sin(x))^2 = \frac{1}{2}(1 - \cos(2x))$.)

This is

$$J = 2\pi \int_{-\infty}^{\infty} \hat{f}(\lambda)^2 d\lambda = 2\pi \|\hat{f}\|_{L_2}^2$$

From Theorem 2.12 we have

$$\|\hat{f}\|_{L_2} = \|f\|_{L_2}$$

2/3

$$\begin{aligned}
S_0 \quad J &= 2\pi \|f\|_{L_2} = 2\pi \int_{-\infty}^{\infty} e^{-2|x|} \sin^2(x) dx \\
&= \frac{2\pi}{2} \left(\int_{-\infty}^{\infty} e^{-2|x|} dx - \int_{-\infty}^{\infty} e^{-2|x|} \cos(2x) dx \right) \\
&= \pi \left(2 \int_0^{\infty} e^{-2|x|} dx - 2 \int_0^{\infty} e^{-2|x|} \cos(2x) dx \right) \\
&= \pi \left(2 \left[\frac{e^{-2x}}{-2} \right]_0^{\infty} - 2 \left[\frac{e^{-2x} \sin(2x) - \cos(2x)}{4} \right]_0^{\infty} \right) \\
&= \pi \left(2 \cdot \frac{1}{2} - 2 \cdot \frac{1}{4} \right) \\
&= \frac{\pi}{2}
\end{aligned}$$

3/3